Input Layer
Multi-scale CNN
Feature Encoder
Shared Temporal Encoder
x_1 x_2 x_3 x_4 x_5 ⋮ x_T
$\mathbf{x} \in \mathbb{R}^{B\times1\times T},\ T = 1024$
Raw aggregate · 8 s/sample
Normalization $\hat{x}_t = \frac{x_t-\mu_{\text{train}}}{\sigma_{\text{train}}}$
$\mathbf{k = 3}$
$\mathbf{k = 5}$
$\mathbf{k = 9}$
Conv1D 1→16 · k∈{3, 5, 9} · s=1 · same padding · IBN + ReLU
Each output: $\mathbb{R}^{B\times16\times T}$
Concat $\mathbb{R}^{B\times48\times T}$
1x1 Conv
48 to 32
s=1, p=0
IBN + ReLU
+ 1x1 skip
64 ch
Conv1D 32 to 64
k=5, s=1, p=2
IBN + ReLU
128 ch
Conv1D 64 to 128
k=5, s=1, p=2
IBN + ReLU
$\mathbb{R}^{B\times128\times T}$
Residual TCN unit
$\mathbf{z}_l$ 128 ch
Dilated Conv1D
128 to 128
$k = 9,\ d = d_l,\ p = 4d_l$
BN
ReLU
Dropout 0.25
$\oplus$
$\mathbf{z}_{l+1}$ 128 ch
identity
$z_{l+1} = z_l + \text{Dropout}\{\text{ReLU}[\text{BN}(\text{Conv1D}_{9,d_l}(z_l))]\}$
Five sequential units; dilation schedule
TCN 1 $d_1 = 1,\ p_1 = 4$
TCN 2 $d_2 = 2,\ p_2 = 8$
TCN 3 $d_3 = 4,\ p_3 = 16$
TCN 4 $d_4 = 8,\ p_4 = 32$
TCN 5 $d_5 = 16,\ p_5 = 64$
k=9; stride=1; T=1024 throughout
$\mathbf{H}$ $\mathbb{R}^{B\times128\times T}$
N=5 Appliance Decoders
Cross-Appliance Relation Attention
N=5 Power-State Output Heads
Appliance i decoder
Channel gate
H 128 ch
1x1 Conv 128 to 32
ReLU
1x1 Conv 32 to 128
σ
$\odot$
$\tilde{\mathbf{H}}_i$
Local residual decoder
$\tilde{\mathbf{H}}_i$
Conv1D 128 to 128; k=3 BN · ReLU
Conv1D 128 to 128; k=3 BN · ReLU
$\oplus$
Dropout 0.25
$\mathbf{f}_i$ $\mathbb{R}^{B\times128\times T}$
$\mathbf{F}$ $\mathbb{R}^{B\times T\times5\times128}$
$\mathbf{Q} = \mathbf{W}_Q\mathbf{F}$ 1x1 Conv: 128 to 16
$\mathbf{K} = \mathbf{W}_K\mathbf{F}$ 1x1 Conv: 128 to 16
$\mathbf{V} = \mathbf{W}_V\mathbf{F}$ 1x1 Conv: 128 to 16
$\otimes$
$\mathbf{QK}^\top$
Scale $\frac{1}{\sqrt{16}}$
Softmax across appliances
$\mathbf{A}(t)$ 5×5
$\otimes$
$\mathbf{A}(t)\mathbf{V}$
V
$A(t) = \text{softmax}[Q(t)K(t)^{\mathrm{T}} / \sqrt{16}]$
Message projection 1x1 Conv: 16 to 128
Message gate 1x1 Conv: 256 to 128 Sigmoid
$\oplus$
Residual exchange
$f_i^{\text{out}} = f_i + 0.25\, g_i \odot m_i$
Appliance *i* output head
$f_i(t)$
128 channels
Power FC
Fully connected: 128 → 1
$y_i^{\text{raw}}(t)$
State FC
Fully connected: 128 → 1
$s_i(t)$
σ
$p_i(t)$
$1 - p_i(t)$
y_i^{off}
$\odot$
$\odot$
$\oplus$
Soft power gate
$y_i^{\text{gate}}(t) = p_i(t)y_i^{\text{raw}}(t) + [1 - p_i(t)]y_i^{\text{off}}$
Power: $\mathbb{R}^{B\times T\times5}$ $y_i^{\text{gate}}(t)$
State: $\mathbb{R}^{B\times T\times5}$ $p_i(t)$